Samukelo Mngadi

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Profile

Business Analyst who turns business problems into requirements, and requirements into working products. I gather requirements, map processes, and prototype AI-assisted solutions so stakeholders can react to something real before development starts.

Experience

Business Analyst Intern, MICT SETA

March 2025 - March 2026

- Elicited, documented and validated business requirements; linked stakeholders and developers.
- Wrote user stories with positive and negative acceptance criteria; turned requirements into development tasks.
- Mapped processes and system flows; built ERDs and data models.
- Designed wireframes in Figma and Visily; redesigned features after stakeholder feedback.
- Tested developer builds (staging, UAT, demos), reported defects, retested fixes, helped plan UAT.
- Scrum and Kanban: sprint planning, prioritisation, roadmaps, project plans; Microsoft Planner.

Projects

Tower: airline complaint triage assistant

Reads each customer complaint, sorts it by reason and urgency, drafts a reply with Gemini, and a person approves every reply before the customer sees it. Agents work a queue sorted by urgency and can escalate to a team lead, who can approve or return it. Customers get a reference and a status page where they see approved replies and can write back. Several airlines share one app, each with its own staff, complaints and activity log, managed by four roles (platform admin, airline admin, team lead, support agent). The reason classifier is right 62.7% of the time on unseen airline tweets against 31.7% for guessing, and 79% when at least 50% sure; less certain complaints are flagged for a manual read. Responsible AI controls: masked personal details, rules against promising refunds or compensation, an AI usage limit and an activity log. Prototype: the social feed is a replay of public Kaggle tweets. Built with Python, Streamlit, scikit-learn, Gemini and SQLite.

Live app: airline-complaint-triage-h5gvccmt8vbmwyfhmpc2n3.streamlit.app | Code: github.com/mngadisamu/airline-complaint-triage

Sentiment analysis tool

A web tool that turns a CSV of customer comments, tweets or reviews into a cleaned dataset, a sentiment dashboard and a downloadable report, keeping a log of every cleaning change. Tested on 14,604 airline tweets from Kaggle, it agreed with the dataset's labels 55.0% of the time at first and 65.9% after tuning for complaint language. Python, pandas, VADER, Streamlit.

Live app: samukelo-sentiment-analysis-tool.streamlit.app | Code: github.com/mngadisamu/sentiment-analysis-tool

AI business analyst assistant

BA Assistant Studio turns a raw stakeholder requirement into a requirement analysis, user story, Given/When/Then acceptance criteria, test cases, clarification questions and a quality check. It is driven by a master prompt that sets its role, outputs and rules, and has a prompt engineering mode. The analyst stays responsible for reviewing everything it produces. React, TypeScript, Gemini API.

Code: github.com/mngadisamu/ai-business-analyst-assistant

OOTD AI: fashion stylist chatbot

A chatbot that builds outfits from your digital wardrobe, with an inspiration hub, saved outfits and wear logs. I defined the features and flows, then prototyped it with Google AI Studio and Gemini (React, TypeScript).

Code: github.com/mngadisamu/ootd-ai-chatbot

Case management system

Gathered requirements, wrote user stories and acceptance criteria, tested features, logged bugs and drew up a sprint roadmap. Redesigned a vague dashboard requirement after consulting other departments.

Onboarding and verification platform

Analysed requirements for employee, supplier and beneficiary onboarding: identity (KYC/AML), address, criminal, financial, citizenship and qualification checks, plus HR dashboards, leave and performance processes. Researched verification API integrations.

Property management system

Business analysis for workflows and system requirements.

Reflection

The AI was the easy part of Tower. The hard work was everything around it: who may see what, what happens when the model is unsure, how a customer follows up, and how one airline's data stays separate from another's. Measuring the classifier honestly is what led me to design it to suggest and flag rather than decide. As a business analyst, I take the same approach into requirements work: write down the edge cases and limits before building, and show stakeholders something real early.

Skills

Business analysis: requirements gathering, user stories and acceptance criteria, process and system-flow modelling, ERDs, stakeholder communication.

Delivery and tools: Scrum, Kanban, SDLC, UAT and defect management, Figma, Visily, Microsoft Planner.

Building with AI: prompt engineering with Gemini, Google AI Studio, React and TypeScript basics, Python, pandas and Streamlit, text classification with scikit-learn, responsible AI (human approval, confidence flags, guardrails), Git and GitHub.

Contact

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